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Code

Issues

Pull requests

Actions

Projects

Wiki

Security and quality

Insights

Settings

AI-CMOS-MEMS-Academy / module-00-ai-era / README.md



mikelix Add Module 0 AI Needs Sensors learning experience ...

8a0824c · 1 minute ago



Preview

Code

Blame

1182 lines (274 loc) · 5.74 KB



Raw



Module 0: AI Needs Sensors

Building the Physical Intelligence Layer for Artificial Intelligence

<img src="assets/images/module0-ai-needs-sensors-hero.png"

alt="AI Needs Sensors - The Physical Intelligence Layer of AI"

width="100%">

Module 00 AI Era

Topic AI Sensors

Technology

CMOS MEMS

Learning

Level Beginner

Welcome to the AI Sensor Revolution 🚀

AI Has Intelligence.

But AI Needs Sensors to Understand the Real World.

Everyone is talking about:

- ChatGPT
- Generative AI
- Large Language Models
- AI Agents

But one fundamental question remains:

How does AI see, hear, feel, and interact with the physical world?

The answer:

Sensors.

The Missing Physical Intelligence Layer of AI

Today's AI revolution is mainly driven by software intelligence.

However:

AI cannot:

- See the environment
- Measure motion
- Detect pressure
- Understand vibration
- Sense temperature
- Interact with physical systems

without sensors.

The next AI revolution requires:

Physical Intelligence

From Artificial Intelligence to Intelligent Systems

A human intelligence system requires:



Brain

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Eyes

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Ears

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Touch

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Movement

Intelligent machines require:



AI Models

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Sensors

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Actuators

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Computing

=

Intelligent Systems

The AI Data Pipeline

Every intelligent system follows:



Physical World

 ↓

Sensors

 ↓

Data

 ↓

AI Models

 ↓

Decision

 ↓

Action

Better sensors create better AI systems.

Why CMOS MEMS?

MEMS means:

Micro-Electro-Mechanical Systems

MEMS integrates:

Mechanical Structures

\+



Sensors

\+

Actuators

\+

Electronics

\+

Communication

\+

Power

=

Microsystem Intelligence

CMOS MEMS combines:



CMOS Semiconductor Technology

 +

MEMS Mechanical Structures

 +

AI Algorithms

 =

Intelligent Sensor Systems

Why CMOS MEMS Sensors Matter

CMOS MEMS sensors provide:

- Small size
- Low cost
- Low power consumption
- High sensitivity
- Mass manufacturability

They are becoming the sensing foundation for:

🤖 Robotics

Robots need sensors to understand:

- Position
- Motion
- Environment
- Interaction

🚗 Smart Vehicles

Autonomous systems require:

- Accelerometers

- Gyroscopes
- Pressure sensors
- Advanced sensing systems

🏢 Smart Buildings

Intelligent buildings use sensors for:

- Environment monitoring
- Energy optimization
- Human activity detection

🏙️ Smart Cities

Future cities require billions of connected sensors generating intelligent data.

Your AI Learning Team

AI CMOS MEMS Academy combines:

Human Expertise

+

Artificial Intelligence Assistants

🤖 AI CMOS MEMS Agent

Your MEMS engineering assistant.

Ask:



Explain how a CMOS MEMS accelerometer works.

The AI Agent helps you:

- Learn MEMS concepts
- Analyze sensor systems
- Explore applications

- Design engineering solutions

🇹🇼 Prof. Yi-Kuen Lee AI CMOS MEMS GPT

Your AI professor assistant.

Learning approach:



Concept

↓

Physics

↓

Engineering Design

↓

Application

↓

Future Research

Ask:



Teach me MEMS sensing principles

like a university professor.

⚡ AI Analog IC Guru

Your analog and mixed-signal design assistant.

Learn:

- Sensor interface circuits
- Analog front-end design
- ADC requirements
- CMOS sensor electronics

Ask:



```
How do I design an analog interface  
for a MEMS sensor?
```

Hands-on Engineering Challenge

Look at your smartphone.

Ask yourself:

How does your phone know orientation?

Answer:

MEMS accelerometer and gyroscope.

How does your phone detect sound?

Answer:

MEMS microphone.

How does your phone become intelligent?

Answer:



```
Sensors
```



Data



Algorithms



AI Applications

Mini Project 0

Design Your First Intelligent Sensor System

Choose one application:

Project A: Smart Robot

Design:



Robot



MEMS Sensor



Data Processing



AI Decision

Project B: Smart Building

Design:



Environment

↓

Sensor

↓

AI Analysis

↓

Intelligent Control

Learning Philosophy

This academy follows:

Learn → Build → Measure → Analyze → Create

Engineering learning:



Theory

↓

Hardware



Measurement



Data



AI Intelligence

What You Will Learn Next

Module 1: MEMS Fundamentals

You will learn:

- MEMS principles
- Scaling effects
- Sensors and actuators
- Micro-system architecture

Start Your AI Sensor Journey 🚀

The future of AI is not only algorithms.

The future of AI is:

Intelligent Systems That Can Sense the World.

Welcome to AI CMOS MEMS Academy.

